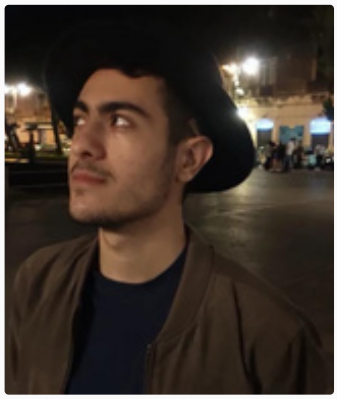


# Claudio Giusti

Software Engineer

**claudiogiusti07@gmail.com · https://claudiunderthehood.github.io**

## Professional Summary

Software Engineer with a strong background in hardware and software development. Experienced in aerospace hardware/software systems and user interface design as well as independent software development as a Freelancer. Passionate about computer science research, specializing in multimedia forensics and Artificial Intelligence with first-author Q1 journal publications. Currently pursuing a Master's degree in Computer Science.

## Work Experience

### FREELANCE SOFTWARE ENGINEER

**Self-Employed | Italy | 2021 - Present**

- Design and deliver custom software solutions and technical consulting for several customers.
- Develop software architectures, data processing pipelines, and code optimization strategies.
- Manage the full software development lifecycle independently, from requirements gathering to final deployment.

### HARDWARE & SOFTWARE DEVELOPER

**ItalSpazio | Catania, Italy | 2018 - 2021**

- Developed hardware and firmware solutions for satellite devices and communication hardware.
- Designed and built intuitive user interfaces (UI) to streamline user interaction with satellite systems.
- Diagnosed and resolved complex hardware-software integration issues.

## Education

**M.Sc. in Computer Science |** University of Catania (Oct 2026 - Present)

**B.Sc. in Computer Science |** University of Catania (Oct 2020 - July 2026)

**High School Diploma in Electronic Engineering and Robotics |** ITIS G. Marconi (2015 - 2020)

## Languages

- Italian: Native
- English:  Proficient / Native-level (Cambridge C2 Certification)
- Chinese: Elementary (HSK 2 Certification)
- Spanish: Beginner (A1 Level)

## Scientific Publications

- **Proto-LeakNet: Towards signal-leak aware attribution in synthetic human face imagery**
  *C. Giusti, L. Guarnera, S. Battiato - Computer Vision and Image Understanding (Elsevier), Vol. 270, 2026 (Q1 Journal)*
- **Fraud is not just rarity: A causal prototype attention approach to realistic synthetic oversampling**
  *C. Giusti, L. Guarnera, M. Casu, S. Battiato - Knowledge-Based Systems (Elsevier), Vol. 350, 2026 (Q1 Journal)*

## Core Skills

**Software Engineering & Research**

- Software Architecture & System Design
- Machine Learning & Computer Vision
- Deepfake Detection & Synthetic Media Forensics
- Explainable & Interpretable AI
- Generative AI

**Technical Tools & Languages**

- **Languages:** Python, C/C++, Java, C#
- **Frameworks:** Pytorch, Tensorflow, OpenCV
- **Tools:** GIT, Docker, Linux

**Soft-Skills**

- Research-Oriented Problem Solving
- Self-Management & Adaptability
- Collaborative Academic & Industrial Teamwork

## Portfolio & Open-Source Projects

- **Proto-Leaknet (GitHub)**
- **VAEGAN-CPAC (Github)**
- **TetrisRLC (Github)**
- **ESP32DeautherTool (Github)**
- **Traction_Control (Github)**
- **Github**

## Interests

- **AI Research & Open-Source:** Tracking state-of-the-art advances in Deep Learning, Computer Vision, and Cybersecurity
- **Embedded Systems & Electronics:** Designing and prototyping custom micro-circuits and IoT hobby projects.

---

*I hereby authorize the processing of my personal data in accordance with the GDPR (EU 2016/679).*